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di Torino

FONDAZIONE
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PASSION FOR INNOVATION

Multimodal Emotion Recognition Using Large Audio-Language Models

ASHKAN SHAFIEI

S342583

PARASTOO ALAVI

S340942

AMIR MASOUD ALMASI

S337006

PROJECT GOALS

1

Design a multimodal AI system to detect emotion

2

Promote human-centered and ethical AI systems



VALUE PROPOSITIONS



IMPROVED
EMOTIONAL WELL-
BEING



DATA-DRIVEN
HEALTH INSIGHTS



ENHANCED
PATIENT SUPPORT
SYSTEMS



RESPONSIBLE &
HUMAN-CENTERED
AI

SUSTAINABLE DEVELOPMENT GOALS



PROBLEM STATEMENT



NOISE IN REAL-WORLD SPEECH

Existing systems fail when audio is noisy or text is ambiguous

FAILURE TO CAPTURE SUBTLE EMOTIONS

Single-modality models cannot capture complex emotional cues

DATASET SHIFT & GENERALIZATION ISSUES

Datasets differ widely, causing generalization issues

HYPOTHESES



NEED FOR EMOTION-SPECIFIC TRAINING

General-purpose audio-language models will perform poorly in recognizing emotions in speech without task-specific training.



QA VS MULTIMODAL TRAINING

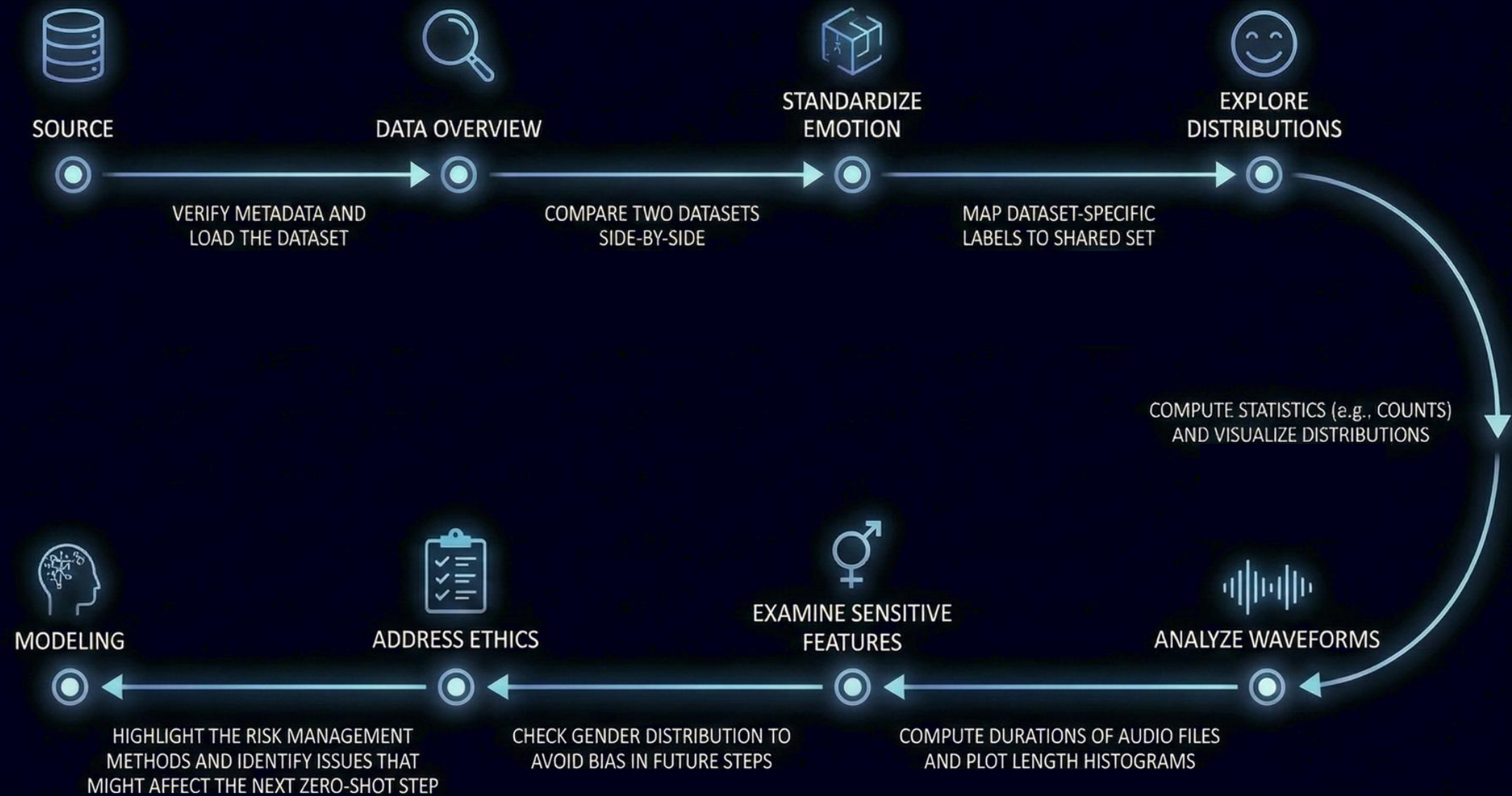
While transcripts contribute only marginal improvements because emotion cues are mostly acoustic, transitioning from QA-only training to emotion-focused multimodal training still results in better performance.



LORA FINE-TUNING BENEFITS

PEFT/LoRA fine-tuning on a balanced dataset (ESD) will significantly improve recognition of high-arousal emotions such as Angry and Happy.

DATA EXPLORATION ROADMAP

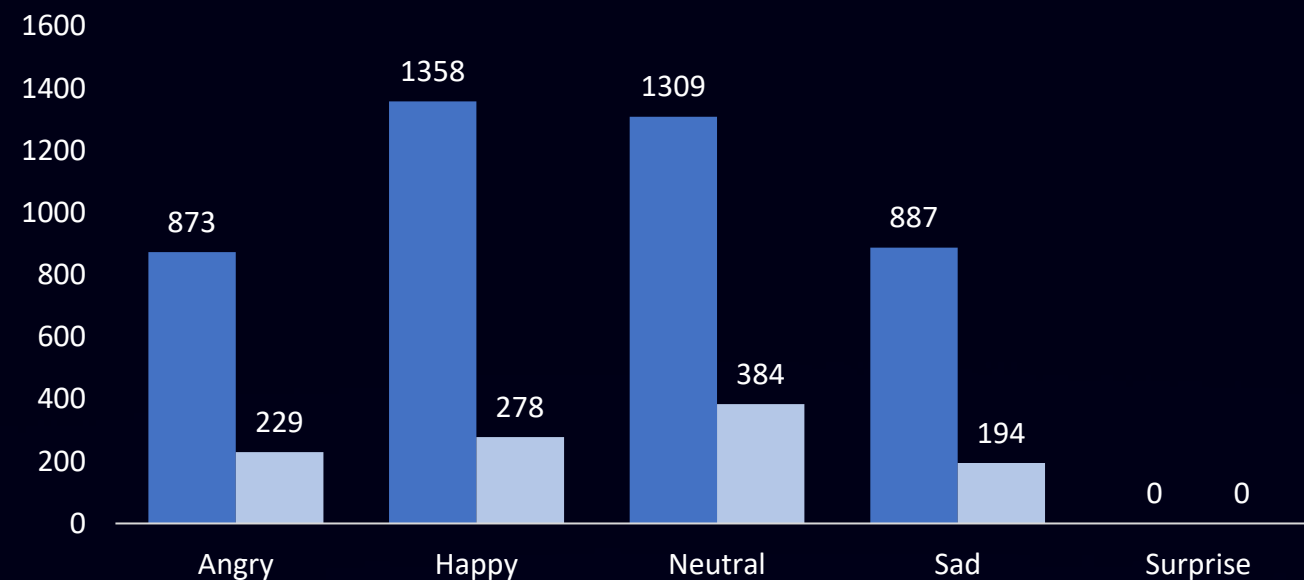


DATA SOURCE

	 ESD	 IEMOCAP
Language	English (This subset is used) + Chinese	English
Emotion Classes	5 (Angry, Happy, Sad, Neutral, Surprise)	4 (Angry, Happy/Excited, Sad, Neutral)
Class Balance	Balanced	Imbalanced
Speakers	20 speakers	10 actors
Recording Type	Read speech (controlled environment)	Natural dialogues
Audio Style	Clear and consistent	More expressive and varied
Metadata	Limited (no gender/age)	Richer metadata (speaker, gender)
Dataset Size	17500 samples (15750 Tr + 1750 Te)	5531 samples (4446 Tr + 1085 Te)
Use In Our Project	Good for fine-tuning (PEFT)	Good for zero-shot testing

EMOTION LABEL DISTRIBUTION: ESD VS IEMOCAP

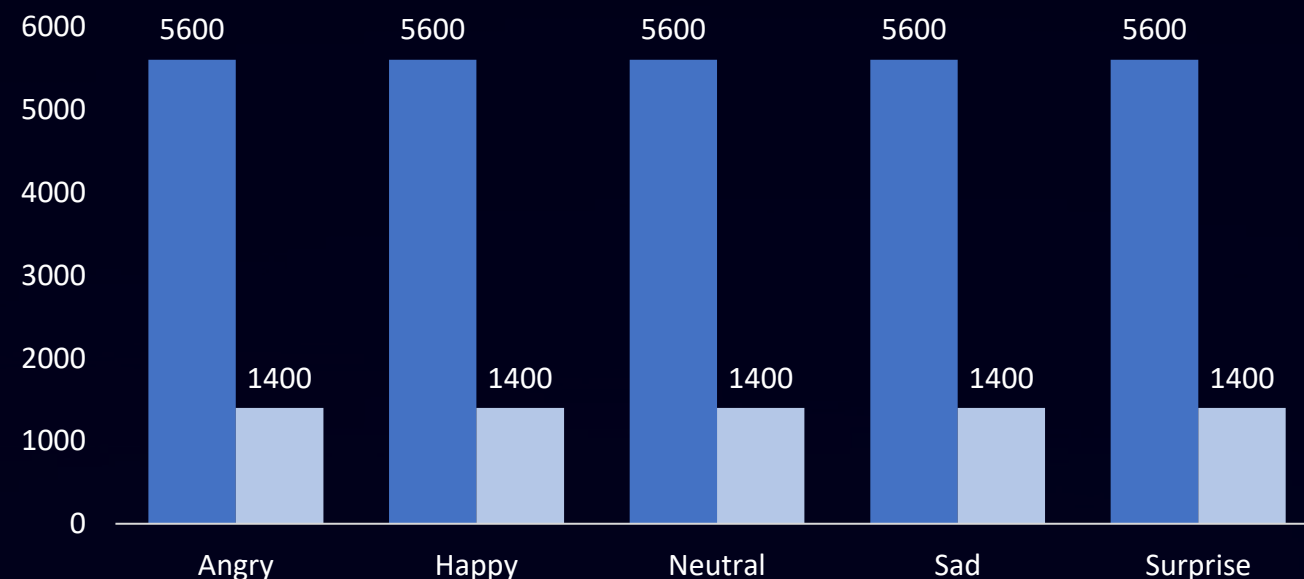
IEMOCAP Label Distribution



IEMOCAP exhibits strong class imbalance

- Emotion frequencies vary significantly:
 - Neutral appears most often
 - Surprise does not exist in IEMOCAP
 - Angry and Sad are the least represented
- Realistic but imbalanced → more difficult for zero-shot generalization.

ESD Label Distribution



ESD shows completely balanced emotion distribution

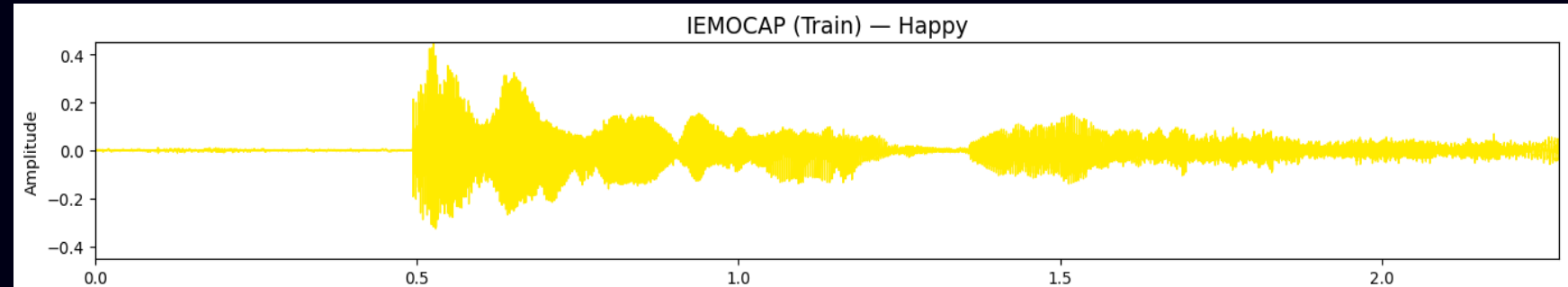
- Perfectly balanced in train and test
- Designed dataset with controlled recordings
- Ideal for training and PEFT fine-tuning

WAVEFORM ACROSS EMOTIONS



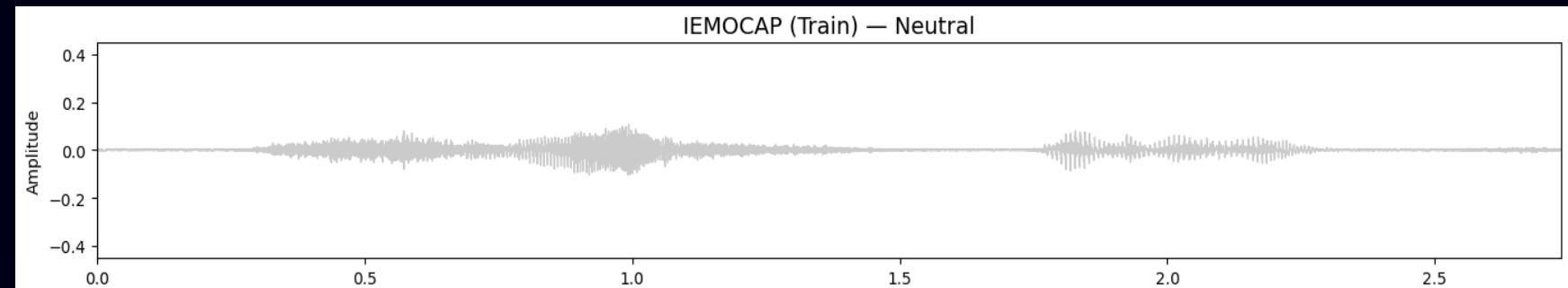
Happy

- Amplitude variations and frequent peaks



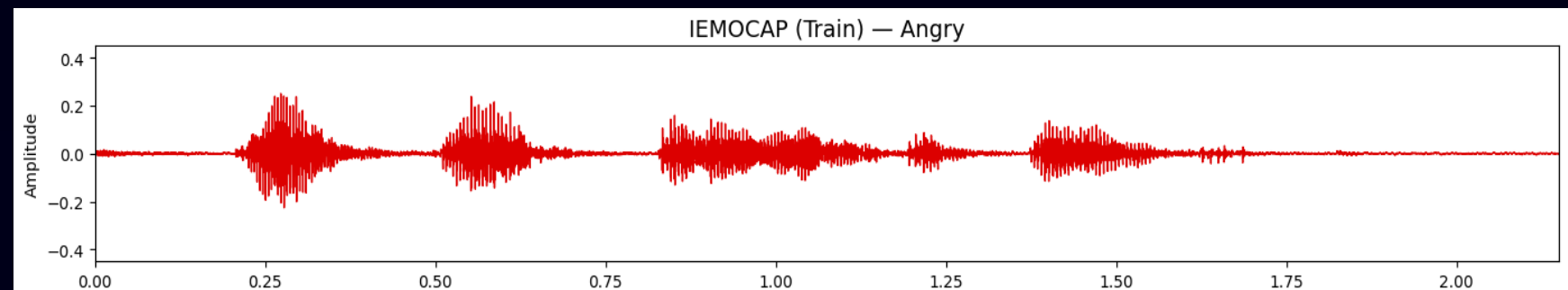
Neutral

- Low amplitude, flat contour, steady and calm



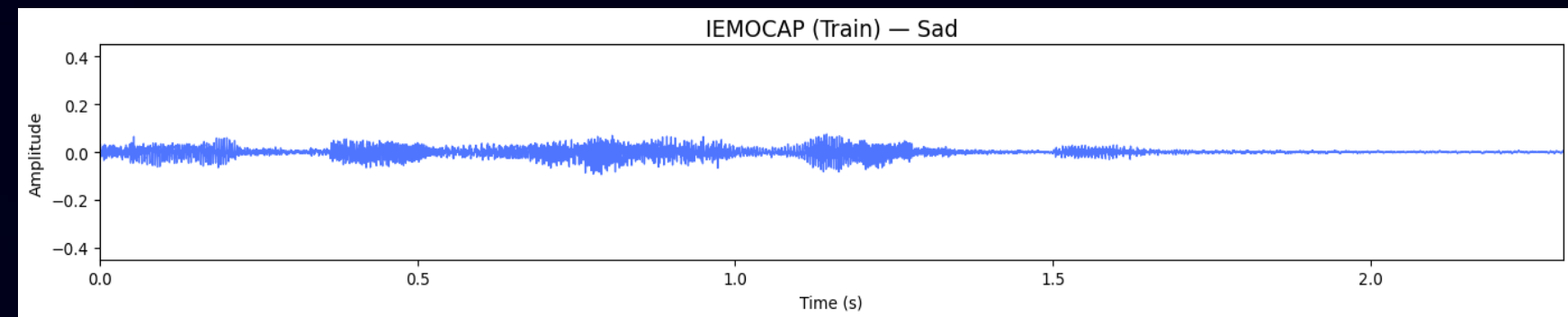
Angry

- Sharp, irregular spikes with high intensity



Sad

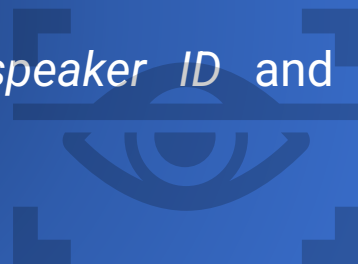
- Low intensity, smooth and longer contours



DATA PRIVACY

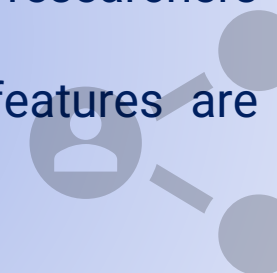
NO PERSONALLY IDENTIFIABLE INFORMATION

- Audio in both datasets contains only speech, no names or private identifiers
- No demographic fields beyond *speaker ID* and *emotion label*



STRICT NON-SHARING OF RAW AUDIO

- Raw .wav files remain local to the researchers and are never uploaded to GitHub
- Only preprocessed, non-reversible features are stored in the repository



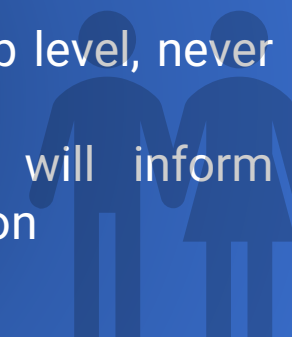
METADATA IS USED FOR FAIRNESS EVALUATION

- Gender inferred from filenames solely for bias and fairness analysis, not for profiling
- No attempt is made to reconstruct or identify speakers.



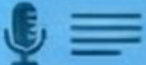
ETHICAL HANDLING OF SENSITIVE ATTRIBUTES

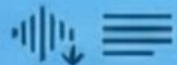
- Sensitive analyses reported at group level, never at individual level
- Any demographic bias identified will inform model evaluation—not personalization



ZERO-SHOT EMOTION ANALYSIS APPROACHES

Voxtral Mini 3B

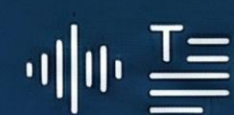
 **Unified Audio and Text Model**
Handles both modalities in a single model.

 **Whisper-based Encoder**
Uses Whisper Large v3 with 4x downsampling.

 **32k Token Context**
Supports 30-40 min audio.

 **Designed for Transcription**
Optimized for semantic audio understanding.

Qwen2 Audio 7B



Audio Enabled Qwen2 Model
7B parameters, multimodal capabilities.



Flexible Input
Accepts audio or audio and text directly.



Stronger Linguistic Reasoning
More capable than Voxtral, but slower with higher GPU cost.

Two Model Pipeline (ASR + Text LLM)



ASR

-  Whisper Large V3 XLS R
-  Stable transcripts, Flexible text reasoning
-  Higher complexity, Error propagation from ASR

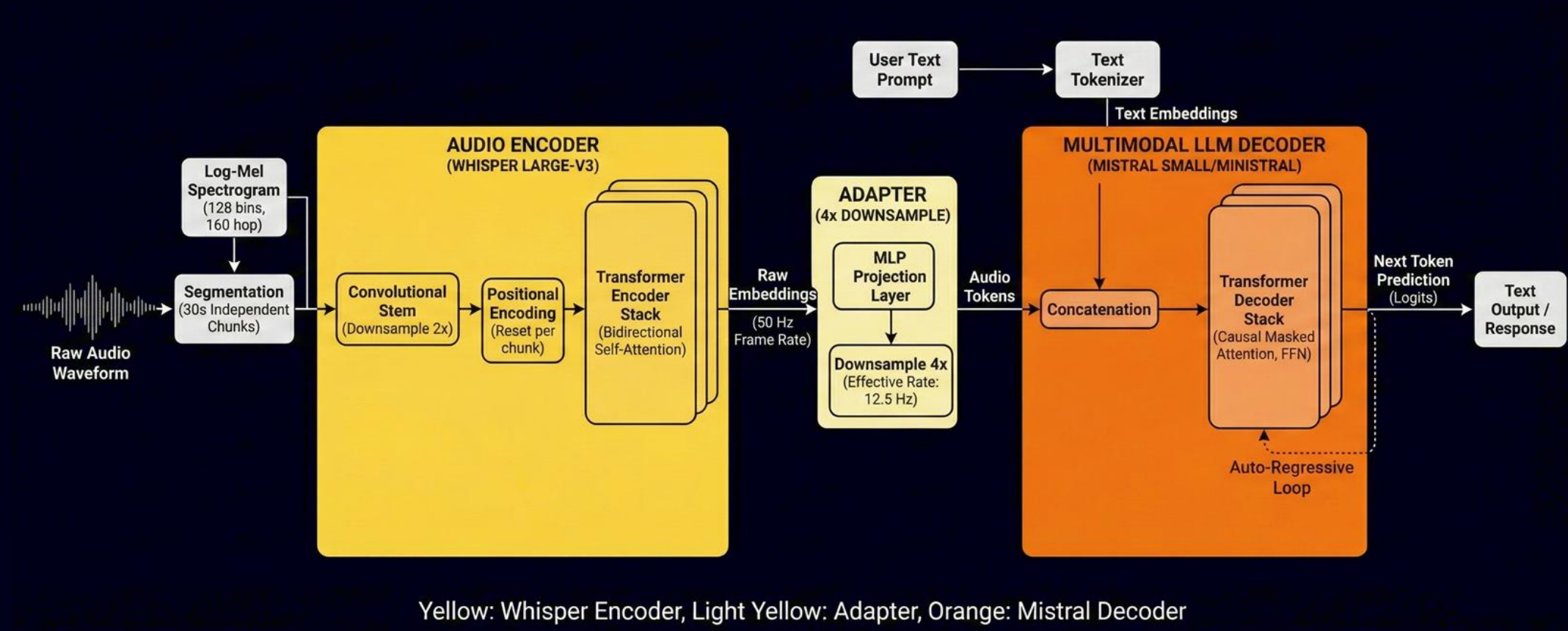


Text LLM

-  Any text-only model (e.g., Qwen2.5 3B)
-  Allows targeted fine-tuning
 - Focus on small emotion datasets

WHY VOXTRAL-MINI-3B?

- One multimodal model removes ASR and LLM error chains, making the system simpler and more reliable.
- The 3B model fits Colab limits, while Qwen Audio 7B is heavier, slower, and harder to run on the full dataset.
- The model combines audio and text natively, so it naturally understands both prosody and meaning together.
- It supports long inputs (up to 32k tokens), allowing us to process full utterances without splitting the audio.

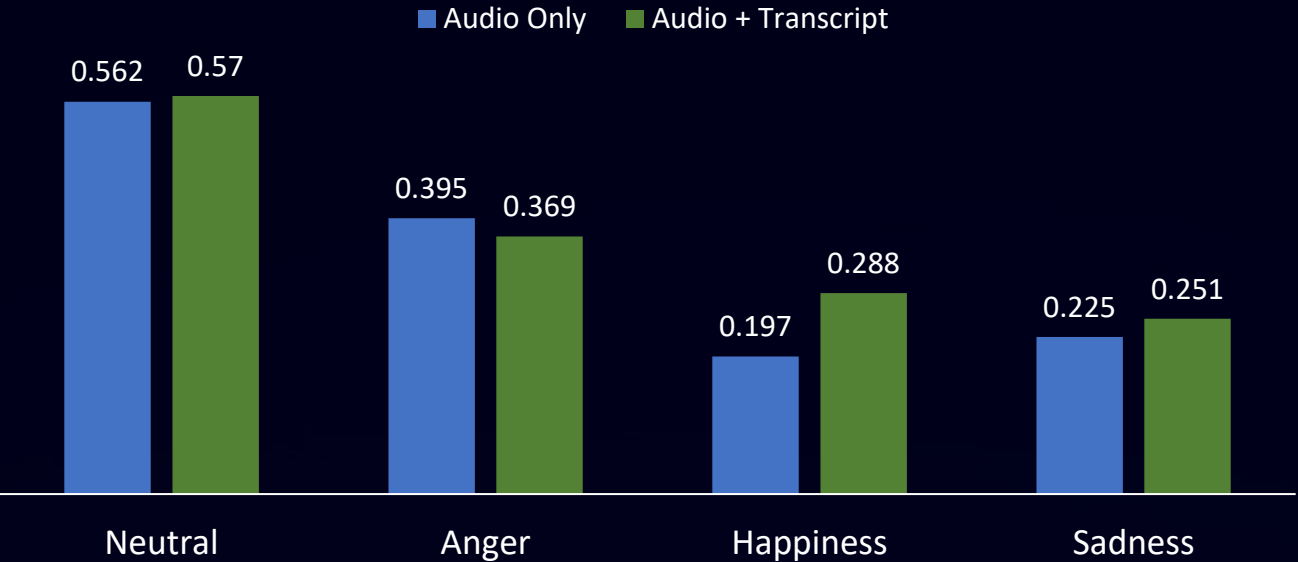


ZERO SHOT PERFORMANCE ON IEMOCAP

	AUDIO ONLY	AUDIO + TRANSCRIPT
Accuracy	0.45	0.46
Balanced accuracy	0.37	0.38
Macro F1	0.34	0.37
Cohen's kappa	0.17	0.20
MCC	0.26	0.29

- Limited Emotion Sensitivity in Zero Shot Mode
- Low Macro-F1 Reflects Poor Class Separation
- The low scores of Cohen's Kappa indicate the model is barely more reliable than a purely random guess.
- The weak values of MCC confirm that the system frequently confuses emotions and lacks strong predictive power.

PER-EMOTION CLASS F1 SCORE

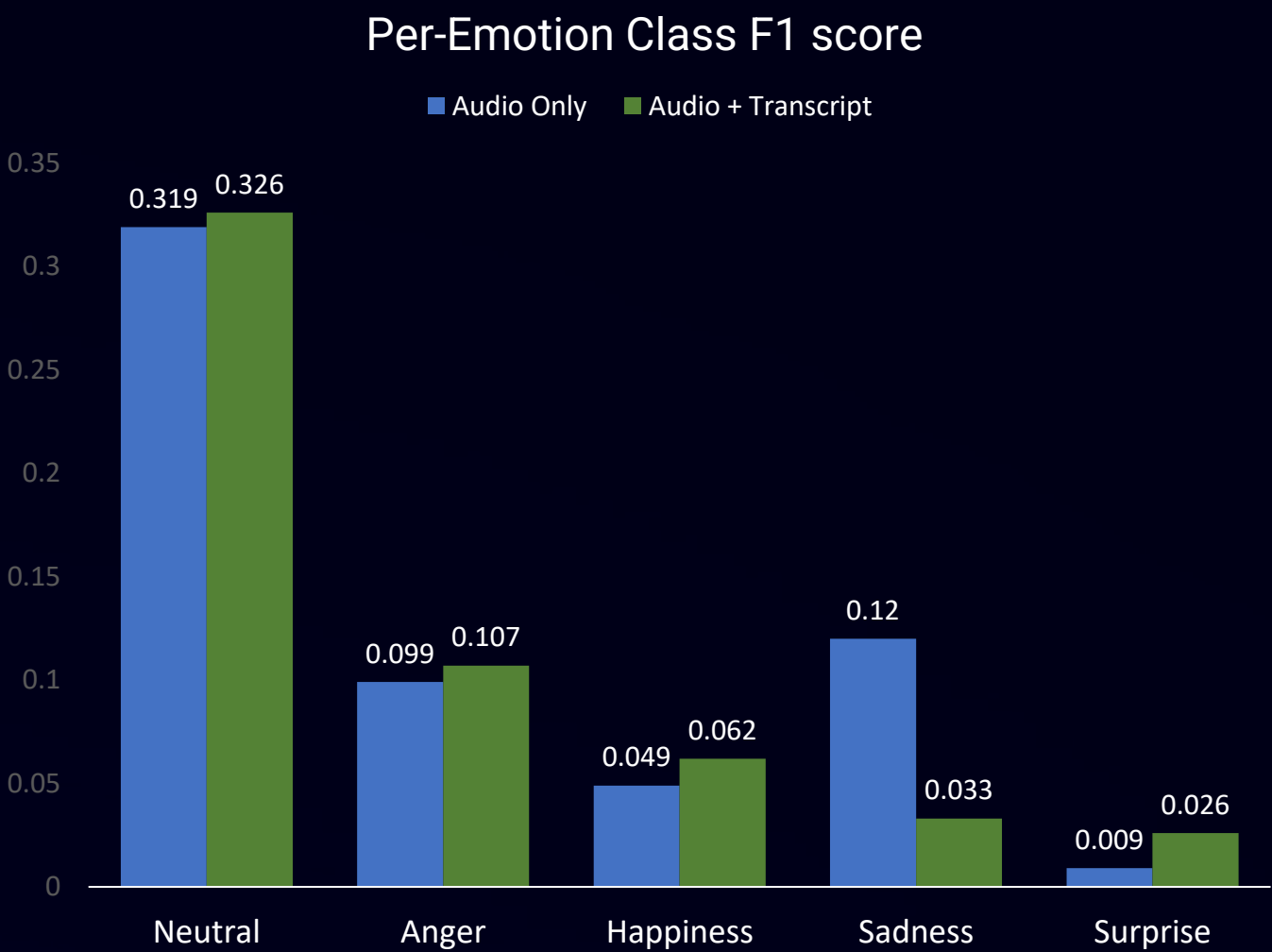


- F1-score Isolates performance for each emotion, preventing the model's bias toward the dominant emotion class from masking its failure on harder categories.
- Neutral Over Prediction Drives Metrics

ZERO SHOT PERFORMANCE ON ESD

As seen in IEMOCAP, zero-shot results on ESD are near random, with only slight gains from adding transcripts. Neutral is the only emotion recognized reasonably well, while others remain poorly classified, confirming that effective emotion recognition requires fine-tuning.

	AUDIO ONLY	AUDIO + TRANSCRIPT
Accuracy	0.195	0.209
Balanced accuracy	0.195	0.209
Macro F1	0.113	0.128
Cohen's kappa	0.004	0.011
MCC	0.007	0.019



AGREEMENT AND DISAGREEMENT PATTERNS

	ESD	IEMOCAP
Both Wrong	4110	568
Both Correct	980	472
A+T Correct	117	30
Audio only correct	43	15

- The dominance of "Both Wrong" proves that pre-trained general knowledge is insufficient for these specific datasets; the model blindly guesses wrong in both modalities.
- In a zero-shot setting, the text modality lacks the domain knowledge to correct the audio errors, resulting in negligible gains from fusion (A+T Correct is barely higher than Audio only correct).

PROJECT PROGRESS AND NEXT STEPS

1. Defined objective, questions, and risk.
2. Explored ESD and IEMOCAP datasets.
3. Ran zero shot with Voxtral (audio/transcript).
4. Analyzed agreement/disagreement.

5. Apply PEFT fine tuning on Voxtral.
6. Re-evaluate hypotheses.
7. Compare zero shot vs PEFT.
8. Prepare updated analysis.

01

02

03

04

05



**THANK
YOU!**